

NITIN YADAV

Computer Science & Engineering Student | Full-Stack & GenAI Developer

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PROFESSIONAL SUMMARY

Highly motivated Computer Science Engineering student with solid experience architecting and deploying production-grade software systems. Adept at developing secure backend APIs, optimizing database schemas, and integrating generative AI interfaces. Possesses deep understanding of Data Structures & Algorithms, Operating Systems, DBMS, and network protocols. Recognized for outstanding speed, technical competence, and modular code structures during software engineering internships.

EDUCATION

VIT Bhopal University

B.Tech in Computer Science Engineering (CSE Core) | **CGPA: 8.83 / 10.0**

2023 – 2027

Bhopal, India

CBSE Class XII (Class 12 - PCM)

Secondary School Education

Pass (77.8%)

Haryana, India

CBSE Class X (Class 10)

High School Education

Pass (89.4%)

Haryana, India

TECHNICAL SKILLS

Programming Languages:

C++, Python, JavaScript, TypeScript, SQL, Java

Web Frontend:

React.js, Next.js, Tailwind CSS, HTML5, CSS3, Framer Motion

Backend & API:

FastAPI, Node.js, Express, RESTful APIs, Django

Databases & Cloud:

PostgreSQL, MongoDB, Supabase, Vercel, Firebase

AI & Machine Learning:

OpenAI APIs, Whisper Speech-to-Text, Prompt Engineering, RAG Architectures

Security & Tools:

Ethical Hacking, OWASP Top 10, Git, Docker, Linux, Linux Bash Shell, Postman

WORK EXPERIENCE

Software Engineering Intern | Frigoglass India Pvt. Ltd.

Nov 2025 – Jan 2026

Gurugram, India

- Collaborated with the Technical (CSE/IT) Department to optimize PostgreSQL database structures and Express/FastAPI REST APIs, reducing average database query latencies by 20%.
- Developed modular, secure, and reusable backend microservices using Python and Django templates, ensuring sanitization of incoming client inputs.
- Contributed core code to the **Nexus** Smart Campus application, integrating responsive web panels with database endpoints.
- Configured local Dockerized testing environments and executed Postman test suites to maintain endpoint security compliance.

Freelance / Independent Full-Stack Developer

2023 – Present

Remote

- Built and deployed 10+ interactive full-stack web applications and AI integration tools for international clients.
- Programmed secure authentication barriers (JWT/OAuth), role-based authorization gates, and payment gateways.
- Configured containerized system deployments using Docker and managed cloud hosting via Vercel.

Event Management Team Member | Data Science Club (DSC)

2024

Bhopal, India

SELECTED PROJECTS

Bhopal Bizlink | *Next.js, Tailwind CSS, PostgreSQL, EPICS Sandbox*

github.com/Nydv01/Bhopal-Bizlink

- Engineered an EPICS coding sandbox simulator allowing students to compile and test code layouts in real-time within an isolated runtime environment.
- Developed a local business matching bridge, linking small enterprise needs with student developers using Next.js and PostgreSQL.
- Configured state persistence and JWT session authorization parameters to protect admin dashboard control panels.

WanderGlow | *Next.js, FastAPI, OpenAI API, Supabase, Tailwind*

github.com/Nydv01/WanderGlow

- Built an AI travel itinerary planner leveraging OpenAI's GPT models to dynamically generate multi-day, personalized travel maps.
- Designed secure vector search database indexing using Supabase PostgreSQL, reducing coordinate retrieval times by 35%.
- Implemented state persistence for itineraries using local storage caching and API query optimization.

DripDynamics | *React, TypeScript, Supabase, Framer Motion, Web Audio*

github.com/Nydv01/vintage-drip

API

- Engineered a comic-book style streetwear storefront using React, TypeScript, and Framer Motion with custom halftone overlays and micro-animations.
- Integrated live audio triggers utilizing Web Audio API and designed a dynamic sticker cursor with particle cursor collision physics.
- Deployed a Supabase backend for inventory control, secure database transactions, and user authentication, reducing page size payloads by 40%.

FOSSEE Socratic Tutor | *Python, FastAPI, OpenAI API, AST Parsing*

github.com/Nydv01/fossee-tutor

- Built an AI tutoring chatbot utilizing AST (Abstract Syntax Tree) code flow validation for student python submissions.
- Engineered system prompt boundaries preventing direct solution exposure, guiding users socratic-style through hints.
- Designed asynchronous FastAPI backend modules with rate-limiting middleware to guard OpenAI API keys against abuse.

Distributed Traffic Routing System | *C++, Multithreading, Socket*

github.com/Nydv01/traffic-routing

Programming

- Developed a multi-threaded routing simulation in C++ implementing mutex locks, condition variables, and thread-safe queues to prevent deadlocks.
- Transmitted simulated routing packets over custom TCP/IP socket network layers between simulated nodes.
- Processed simulated routing packets at 10,000+ packets/sec without any packet collisions or memory leaks.

CERTIFICATIONS

• **Google Cybersecurity Professional Certificate:** Network security, SIEM logs analysis, threat detection, SQL, Linux & risk mitigation (2025)

coursera.org/verify/cybersec

• **Industrial IoT Markets & Security Certificate:** IoT architectures, cyber-physical threats, and protocol security (2025)

coursera.org/verify/iiot-sec

• **Google Generative AI Certification:** LLM alignment, prompt engineering, and model application workflows (2025)

coursera.org/verify/gen-ai

• **Verified Credly Badges:** Systems administration, networking protocols, security logs analysis, and scripting automation.

credly.com/users/nitin

RELEVANT COURSEWORK

Data Structures & Algorithms · Object-Oriented Programming (C++/Java) · Database Management (DBMS) · Operating Systems · Computer Networks · Web Security & Cryptography · Software Engineering